Aditya Rajput

+91 8604672115 | Email | LinkedIn | GitHub | Portfolio

Computer Science undergraduate with hands-on experience building Python and FastAPI backends, integrating LLM-powered applications, and optimizing PostgreSQL queries. Developed full-stack products across legal-tech and healthcare, with practical experience in authentication, Docker-based environments, database design, and API development.

EDUCATION

Pranveer Singh Institute of Technology (PSIT), Kanpur, India **2024 – 2028**
Bachelor of Technology (B.Tech) in Computer Science and Engineering

Delhi Public School Kanpur **85% | 2024**
Intermediate (Class XII)

Delhi Public School Kanpur **87% | 2022**
High School (Class X)

TECHNICAL SKILLS

Languages: Python, C, JavaScript, SQL
Web & Frontend: HTML, CSS, React.js (Vite), Node.js, Express.js, Tailwind CSS
Backend & APIs: FastAPI, Flask, Django, REST APIs, Uvicorn, SQLAlchemy, JWT Authentication, WebSockets
Databases: PostgreSQL (Supabase), MySQL, MongoDB, Redis
AI / ML: LLM Integration (Groq, Grok, Claude API), RAG (LlamaIndex, ChromaDB), Prompt Engineering, Image Classification (ResNet18)
Tools: Git, GitHub, Postman, Docker, Docker Compose, Kubernetes, pytest, Render

PROJECTS

Nyaya AI – AI-Powered Legal Aid Platform | GitHub **2026**
FastAPI, Python, Groq API (llama-3.3-70b), Supabase, React.js, Tailwind CSS

- Built a full-stack AI legal assistant for Indian citizens with bilingual support (English & Hindi) and persistent query history.
- Integrated Groq's llama-3.3-70b LLM via FastAPI to deliver real-time, context-aware legal guidance tailored to Indian law.
- Implemented Supabase Auth (Email + Google OAuth) with PostgreSQL for secure user management and query storage.

FRACTOES – AI-Powered Fracture X-Ray Analysis Platform | GitHub **2026**
FastAPI, Python, SQLAlchemy, Supabase (PostgreSQL & Storage), Grok API, ReportLab

- Sole backend developer on a 5-person team building an AI-assisted fracture detection and radiology reporting web app.
- Designed and deployed the FastAPI backend (Render) with JWT auth, OTP email verification, and Supabase-backed storage.
- Integrated a ResNet18 fracture classifier (~95% accuracy) with Grok-generated radiology reports and ReportLab PDF output.

Nirnay – AI-Powered Policy Decision Auditor | GitHub **2026**
Python, FastAPI, LlamaIndex, ChromaDB, Claude API, PostgreSQL (Supabase), JWT, web3.py

- Built a FastAPI RAG backend that answers policy questions from institution PDFs with cited sources and confidence scores, using LlamaIndex, ChromaDB (one isolated collection per institution) and the Claude API.
- Implemented a rule-based policy engine (approve / flag / escalate), deterministic SHA-256 hashing of every decision record, and a public no-login endpoint that detects tampering.
- Built JWT role-based auth with institution verification gating, and a web3.py service to anchor record hashes on Polygon Amoy (mock mode until contract deployment).

EXPERIENCE

FlyRank – Back-End AI Engineering Intern **Jul 2026 – Oct 2026**
FastAPI, PostgreSQL, Redis, Docker, Supabase Auth, pytest

- Containerized FastAPI, PostgreSQL and Redis with Docker Compose; swapped storage with zero route or service changes.
- Cut query time 27x (2.41 ms to 0.089 ms) with PostgreSQL indexing, benchmarked using EXPLAIN ANALYZE.
- Built Supabase Auth JWT middleware tested with pytest; ran Claude vs. ChatGPT prompt comparisons.

ACHIEVEMENTS & CERTIFICATIONS

- Technical Head, Google Developer Groups on Campus (GDGoC) PSIT.
- Completed Walmart Global Tech "Advanced Software Engineering Job Simulation" via Forage (August 2026) – covering Advanced Data Structures, Software Architecture, Relational Database Design, and Data Munging.
- Certified in Data Analysis with Python by IBM via Coursera.
- Completed Anthropic Academy certifications in Claude 101 and AI Fluency for Students (2026).
- Semifinalist at BuildWithBharat 2.0 National Hackathon.
- Finalist at the BBD National Hackathon.